# SimPaths Online: user guide

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**CeMPA group preview**

SimPaths Online runs the UK model on a remote server, through your browser. SingleRun brings the JAS-mine interface to the web, with interactive charts, input editing and an AI assistant. MultiRun queues experiments and runs configurations in parallel when resources permit.

| Service | Temporary address | Best for |
|---|---|---|
| **SingleRun** | https://13.140.163.1:8443 | Exploring one simulation interactively |
| **MultiRun** | https://13.140.163.1 | Repeated simulations and policy comparisons |

> **This is an experimental alpha preview.** Please explore the services, push their limits and report problems. Experiments may fail, outputs may be incomplete, and access may be interrupted. Do not rely on work finishing or remaining recoverable. Keep important settings, inputs and downloaded results separately.

The supplied datasets are the **public training data in the SimPaths GitHub repository**, including tax/benefit donors. They are for learning and testing; results are **not suitable for substantive research analysis**. You can upload your own inputs, but please use only public or other non-confidential test data during this preview. See the training-data guidance.

SingleRun is public and currently allows **one user at a time**. MultiRun requires an approved email address and a verification code; no account password is needed. Most server capacity is assigned to MultiRun, which can currently run up to six configurations at once, subject to their resource requirements.

MultiRun shares capacity fairly: users without an active job receive priority; otherwise older eligible jobs are considered first, subject to available resources and per-user limits. Running jobs are not interrupted to admit another user's work.

## 1. MultiRun: submit your first experiment

1. **Sign in with your approved email address** and the six-digit code sent to that inbox. Check spam or junk if it has not arrived. Contact the administration if your address needs approval, and keep cookies enabled.
2. Open **New Experiment** and choose **UK training — 20,000 people** as the **Default input dataset**. The **50,000 people** example is also available to every approved user. Neither requires uploads or UKMOD setup.
3. Keep or edit the suggested **Experiment name** (`Policy Comparison N`). For these fixed examples, `popSize` (population size) and `startYear` (first year) are fixed by the dataset. `endYear` (last year) defaults to **2026**; choose **2020** and **one repetition** for a quick first test.
4. Set **Repetitions per configuration**. A configuration is one scenario with its own inputs and settings; repetitions differ by random seed. All configurations share the seed sequence, starting at `randomSeedIfFixed` (default **606**): **606, 607, 608, …**. Two configurations × three repetitions means **six simulations**. Repetitions run sequentially within each configuration; configurations can run alongside one another.
5. Name and edit the first configuration. Use **Duplicate** or **Add another configuration** to create alternatives, then change their settings as required.

6. Choose a reference under **Baseline for later comparisons**, or leave it unset. Choosing one now lets comparisons prepare automatically as configurations finish. You can choose or change the visualisation baseline later without rerunning simulations.
7. Click **Review experiment**. Check the highlighted differences, inputs, seeds, simulation count and runtime/storage allowances. Use **Back to edit** for corrections. Nothing runs yet.
8. Click **Submit to queue**, then open **My Jobs**. Jobs continue after you close the browser or sign out; sign in again to return to them.

**For larger tests**, choose **UK training — configurable population**. MultiRun supports `popSize` up to **100,000** and `endYear` up to **2070**. Its current limits are **100 configurations** and **10 repetitions per configuration**, hence **1,000 planned simulations per experiment**. Admission also depends on queue allowances and storage; these limits do not guarantee every combination will succeed.

### While work is running

**My Jobs** shows queued, running, completed and unsuccessful work. A configuration may wait for input preparation, server resources, storage or a retry. Read its status before resubmitting. Automatic recovery may increase resources or retry an eligible failure within the recorded limits and available capacity; success is not guaranteed.

After **all repetitions** of a configuration finish, **Validating outputs** means the server is checking the completed files before confirming success. These checks do not establish the scientific validity of the results. If they fail, the status is **Output validation needs review**; the simulations are not automatically rerun. Contact the administrator for review. Chart preparation is a separate later stage, with its own status and retry controls.

**Cancel** requests that unfinished work stop. **Review retry** for an eligible failed configuration, or **Retry** for a cancelled one, opens a review. **Confirm retry** queues it again from the beginning with its saved settings and seeds. Cancellation/retry history is retained. Retries share the original attempt and runtime limits: cancelling after an attempt is created spends that attempt and its elapsed time; cancelling before a new attempt starts spends no additional attempt.

**View configurations** compares saved parameters, inputs and seeds, with differences highlighted. The original submitted baseline appears first, and later additions are included. **Copy to New Experiment** creates an editable draft. **Add Configurations** extends the existing experiment without rerunning completed configurations, keeping its model version and seed plan. Its review includes the original baseline as a reference when one was specified.

Completion and problem emails supplement **My Jobs**; check the website for current status. Emails do not contain your simulation data.

## 2. MultiRun: see your aggregate results in the Visualiser

Results open in the SimPaths Policy Impacts Visualiser, developed by Reese Green at the University of Glasgow. SimPaths Online aggregates simulation outputs on the server before sending chart data to the Visualiser. For training/provider datasets, users receive aggregate results without access to the underlying microdata. Downloads from your own inputs are described in section 3.

1. In **My Jobs**, choose **View Results**. You can view ready charts while other configurations are still running or have failed.
2. Keep the existing **Baseline for visualisation**, choose a completed configuration, or select **No baseline — view levels only**. This changes the visualisation reference,

not the submitted experiment, and reruns no simulations. Selecting a baseline alone starts no calculations.

3. Click **Prepare comparisons** to request the calculations in advance, or **Open Visualiser** to request them and open available charts. Level charts can be viewed while comparisons are being prepared; you do not need to wait for every comparison to finish.
4. Choose outcomes, population groups and scenarios. Scenario buttons show/hide alternatives; hover over charts to see values. **Connect Data** identifies the source and baseline. New charts and comparisons appear automatically in the same view; select newly available alternatives as required.

Initial chart preparation can take minutes for small outputs and over an hour for large, long-running experiments. The current service allows **up to three hours per preparation**, including source checks, aggregation and comparison calculations; queueing can add to the wait. This is a time limit, not an estimated completion time. If Results says **Chart preparation needs retry**, request preparation again there. Compatible summaries from completed repetitions are reused; simulations are not rerun. Ready configurations can be viewed while the others prepare.

Changing the baseline also reuses retained per-run summaries where available. Earlier comparisons are cached separately and can be reused within permission, retention and storage limits.

Differences use **matching random seeds** between baseline and alternative. Variation in the paired differences across repetitions informs the uncertainty estimates; one repetition cannot estimate that variation. After your first small test, try three or more repetitions.

### Save or reopen charts

**Download comparison data** in **Connect Data** saves an **aggregate chart-data ZIP** containing all configurations ready when downloading starts, including hidden ones. Extract it, choose **Open saved comparison**, and select the folder containing `manifest.json`. This is a fixed snapshot: it gains no later completions. Chart export controls are also available. Opening local folders needs a supported directory picker; Chromium has been tested. Viewing online charts does not require that picker.

**Visualise Locally Saved Data** processes compatible raw microdata CSV folders on your computer. Use **Getting Started → View folder structure** to organise Baseline and Scenario folders. Local files are processed entirely in your browser and are not uploaded. **View Online Results** returns from saved/local data to the online experiment. For stale online charts, try **Troubleshooting → Reload charts from server**; server results and downloaded copies stay unchanged.

## 3. MultiRun: inputs, downloads and useful shortcuts

| Page or control | What it does |
|---|---|
| **Create Input Dataset** | Upload compatible population CSV and UKMOD donor files; choose the model release, any replacement workbooks, and policy schedule. Click a workbook name to edit it in the integrated spreadsheet editor. Review and submit preparation; uploading alone does not prepare a dataset. |
| **My Datasets** | Track preparation and reuse inputs. Configurations using pending inputs wait until preparation succeeds. |
| **Storage** | Manage **Uploads**, **Prepared Input Datasets** and **Output Files**. Sections collapse and remember your choices. Check deletion dates and dependency warnings before freeing space. |

| Page or control | What it does |
|---|---|
| **Parameter sweep** | Generate combinations from a starting configuration. Boolean parameters include true and false automatically; numeric parameters accept lists or ranges. Use **Preview sweep**, then **Add generated configurations**. Review and submit the experiment to queue it. |
| **Import or export YAML settings** | Save or refill an experiment form. YAML contains settings and dataset references, not the input data. Importing starts no jobs. |

The web services **read precomputed UKMOD outputs; they do not run UKMOD**. A changed policy schedule needs suitable donor files. For your own inputs, follow the preparation form's format guidance. An experiment's datasets must use a compatible, common model version.

Service updates do not change submitted experiments: running and queued jobs keep their original model version, inputs, parameters and random seeds. Existing prepared datasets are not automatically rebuilt.

**Training/provider datasets are aggregate-only in MultiRun:** you can view and save permitted charts, but cannot download raw inputs or individual-level outputs. For your own uploaded inputs, permitted **Download CSV output (.zip)** and comparison downloads are available; **Include input dataset** optionally bundles retained inputs. **Download a comparison** appears only for eligible configurations. The Visualiser's aggregate ZIP is a separate download, also available for training examples.

### Free space while keeping Visualiser data

Use **Storage → Output Files** to remove large raw outputs once their Visualiser summaries are ready:

1. Check the configuration's charts in **View Results** and save an aggregate ZIP if you want a copy for longer-term use.
2. Choose **Delete raw outputs, keep Visualiser data** beside that configuration. Wait for the retained-data check and review the space to reclaim and summary retention deadline.
3. Read the permanent-deletion warning, then choose **Confirm raw output deletion**. Check the item's status to confirm removal. Deletion can wait for a backup, download or results processing to release the files.

This option requires compatible saved summaries for **every repetition**. A ready chart alone is insufficient. If the check reports missing summaries or preparation needing a retry, return to Results and prepare the charts first. A failed check leaves the raw files in place and shows an explanation beside the action.

After removal, existing charts remain available until their own expiry. You can choose another baseline and prepare comparisons while the required summaries remain valid. Raw-file downloads are lost permanently, and new variables or calculation versions needing the raw files cannot be prepared. **Deletion does not extend any retention deadline.**

**Delete output files** is the full-deletion choice: it also removes access to dependent hosted Visualiser results. After raw-only removal, that control is named **Delete results and Visualiser data**. Job history and copies you already downloaded are kept.

## 4. SingleRun: run and explore a simulation

Open SingleRun and select a model card. Public training profiles do not require Multi-Run sign-in. **Quick Start — 20,000 people** is the easiest first choice; **Quick Start — 50,000 people** uses a larger prepared population. Both have `startYear` fixed at **2019** and `endYear` defaulting to **2026**, with a maximum of **2026**. Choose **README** inside a session for that profile's instructions.

1. Check the **Parameters** sidebar and output options. For a shorter test, set `endYear` to **2020**. Check the seed settings (`fixRandomSeed` and `randomSeedIfFixed`) and choose export settings before Build.
2. Click **Build** and wait for **Start**. Build sets up the population, model and charts; it can take a while.
3. Click **Start** to run, **Pause** to inspect progress, or **Step** to advance incrementally. Pause may need to wait for the current year's calculations to finish. Charts update as the simulation progresses.
4. Use the controls below to inspect inputs, charts, logs and available data. The **AI** assistant is described in section 5.
5. Download anything you want to keep, then click **Leave** to release the session for another user. Session files are removed when the session is cleaned up.

| Control | Purpose |
|---|---|
| **Parameters / Update Params** | Set up a run. During a simulation, Update Params offers only settings permitted to change at runtime. Locked settings require Reset and another Build. |
| **Input Files** | Browse/download accessible inputs; edit permitted workbooks in the spreadsheet editor or upload replacements before Build. Read-only files stay protected. Workbook edits do not automatically repeat input preparation. |
| **Reset** | Reset for another Build. The selected dataset is kept; earlier run folders may remain in Output. |
| **Speed / Grid / Stack** | Adjust execution delay or chart layout. Speed does not allocate more server CPU. |
| **DB** | Inspect permitted input/output tables with read-only queries. Output tables require `exportToDatabase` to have been enabled before Build. |
| **Output** | Download individual files or **Download Complete Run**. Set CSV/database and `persist…` output options before Build. |
| **Console** | Read progress and diagnostics; expand the pane or download the received log. |

Public training sessions permit detailed-data inspection and downloads. Restricted profiles can disable DB access and detailed exports; use the permissions shown in the session to see what is available.

The **configurable training session** lets you review/rebuild bundled input tables. **User-supplied data** has a startup wizard for compatible population/UKMOD files and a policy schedule, with bundled examples also available. Follow **Review selected actions → Continue** before Build. The user-data profile caps `popSize` at **50,000** and has no fixed 2026 ceiling on `endYear`. Longer horizons still depend on input suitability and resources; long-run capacity is not guaranteed.

### If SingleRun is busy

The **one-session limit covers all four profiles**. **Email me when SingleRun is available** lets you verify an address and request one notification. It **does not reserve a place**; another user may take the slot before you return.

After an explicit **Leave**, notices normally wait five minutes in case the previous user is restarting. Closing the browser without Leave holds the slot until idle expiry, normally after an hour; there is no additional five-minute delay after that cleanup. Notification requests expire after seven days.

The optional **SimPaths news and updates** checkbox is separate and unchecked initially. Without it, your address is used only for the availability service. Updates signups are retained separately for up to a year and have a management/ withdrawal link. Integration with a SimPaths mailing list remains future work.

## 5. SingleRun: use the AI assistant

Click **AI** to open **JAS-miner**, the JAS-mine AI assistant. Choose a provider, enter your own provider API key, and select a model and reasoning level. Provider choices are Gemini, OpenRouter, Anthropic and OpenAI; availability and charges depend on your provider account. The assistant is optional.

Try questions such as:

- “Explain the parameters and assumptions behind this chart.”
- “Compare the available runs: which settings and outcomes changed?”
- “Help me diagnose this Build error. What is confirmed and what is uncertain?”

The assistant can inspect permitted charts, settings, workbooks, diagnostics, outputs and database information, and help explain model code. Its tools follow the session’s data-access restrictions. Check scientific documentation and numerical results when interpreting its answers.

Use **camera** to preview/crop a screenshot, **paperclip** to attach a supported file, **Edit & Retry** after a failed request, and **Download** to save the conversation as Markdown.

**Data access matters.** Detailed-data tools are enabled by default for public training sessions. In the user-data profile they start disabled; the data-access button lets you enable them. Relevant records can then be sent to your AI provider. Disabling access starts a fresh conversation but cannot withdraw information already sent. Text, files and screenshots you attach are shared regardless of that tool setting.

Keys are kept in this browser; **Remember on this device** makes them persist there. SimPaths Online does not store them server-side. OpenAI requests forward the key through the SimPaths Online backend to the provider. Use a trusted device and consider the provider’s terms before sharing data.

## 6. Keep results, manage access and report problems

**SingleRun is temporary working storage.** Download needed files before Leave, session replacement or expiry. Closing the browser does not save a session permanently: idle sessions are normally cleaned up after an hour without browser activity. Each current profile has a **10 GiB** storage allowance. If another Build cannot fit, use **Output → Review storage cleanup**, save needed files and review deletions before confirming.

**MultiRun normally retains successful output files for seven days after completion.** Prepared datasets and uploads also have seven-day policies, with clocks linked to creation/editing and actual processing use. Dependencies protect files needed by unfinished work or retained datasets. Viewing, selecting or downloading an item does **not** renew its lifetime. Check dates and protection messages in **Storage**, **My Datasets** and Results; deletion warnings normally arrive 96 and 24 hours before removal. Shared training datasets do not expire under this user-file policy.

Aggregate charts and reusable summaries have separate seven-day retention settings: chart lifetime starts when the charts become ready; summary protection is based on simulation completion. Use the deadlines shown in the service, especially for comparisons whose configurations finished at different times. Storage pressure can reclaim eligible charts unused for 72 hours. These can be regenerated on request while the required summaries or raw outputs remain available. Save the aggregate ZIP for longer-term viewing. Use the raw-only deletion action above to keep eligible Visualiser data when freeing space; full deletion also removes hosted results. Server backups support service recovery, not a promise to recover an individual's deleted or expired files.

For time-limited MultiRun access, renewal reminders arrive **one calendar month** and **seven days** before expiry. Ask the administration to renew it. At ordinary expiry, access and new starts stop; an already admitted attempt may finish within its original deadline and at most 48 hours after account expiry. Explicit revocation stops affected work promptly. Neither event immediately deletes your data: normal retention applies, and renewed access reaches only files that remain.

**If something fails:** read the status or Console and record the time, service, experiment/configuration or session reference, settings and steps that led to it. Send a screenshot or useful log extract to the administration, removing private data and keys. A maintenance page usually means a short update; retry later. Avoid duplicate submissions while checking an existing job.

Your testing, feedback and feature suggestions are valued. Tell us what confused you, what broke and what would help. Capacity, resource allocations, retention and interface choices can be adjusted. More SingleRun capacity could use resources currently assigned to MultiRun or be supported by a server upgrade.